CONCEPT PROTOTYPE · SYNTHETIC DATA

MedIntel Concept Prototype

How to use it

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> **In short:** double-click **MedIntel_Prototype.html**, switch the browser to full screen and click **Guided demo**. No internet connection or installation is needed.

1 What this is

MedIntel_Prototype.html is a clickable concept prototype of MedIntel, the multi-agent clinical decision support framework described in our Project Stage I report. It shows how the system is designed to look and behave once Project Stage II builds it: records are validated, four agents analyze each case in parallel, a fusion step scores the decision with Equation (4.2), and a clinician makes the final call.

Everything runs inside the web browser from that one file. Every patient is synthetic, and every screen carries the label "Concept prototype · synthetic data".

2 Quick start

1. Copy the folder to the laptop you will present from. If you received a .zip file, extract it first (on Windows, right-click it and choose Extract All; on a Mac, double-click it).
2. Double-click **MedIntel_Prototype.html**. It opens in your default browser. Use Google Chrome, Microsoft Edge or Safari; if it opens in another browser, right-click the file and choose Open with.
3. Switch to full screen: `F11` on Windows, `Ctrl` + `Cmd` + `F` on a Mac. Keep the browser zoom at 100% (`Ctrl` + `0` resets it, or `Cmd` + `0` on a Mac).
4. Click **Guided demo** at the top right and follow the steps (Section 4).

To start again from a clean state at any time, reload the page (`F5`, or `Cmd` + `R` on a Mac). Starting the guided demo also resets the demo data.

3 What is in this folder

File	What it is
MedIntel_Prototype.html	The prototype. Double-click it to open it in a browser.
How to Use the MedIntel Prototype.pdf	This guide.
README.txt	A one-page quick start.
sample_data/morning_batch.csv	The ten-record batch used in the demo: eight valid rows and two with errors.
sample_data/boundary_tests.csv	Six more synthetic patients for showing the rules on new data (Section 6).
source_code/	Optional. Only needed if you want to change the prototype (Section 11).

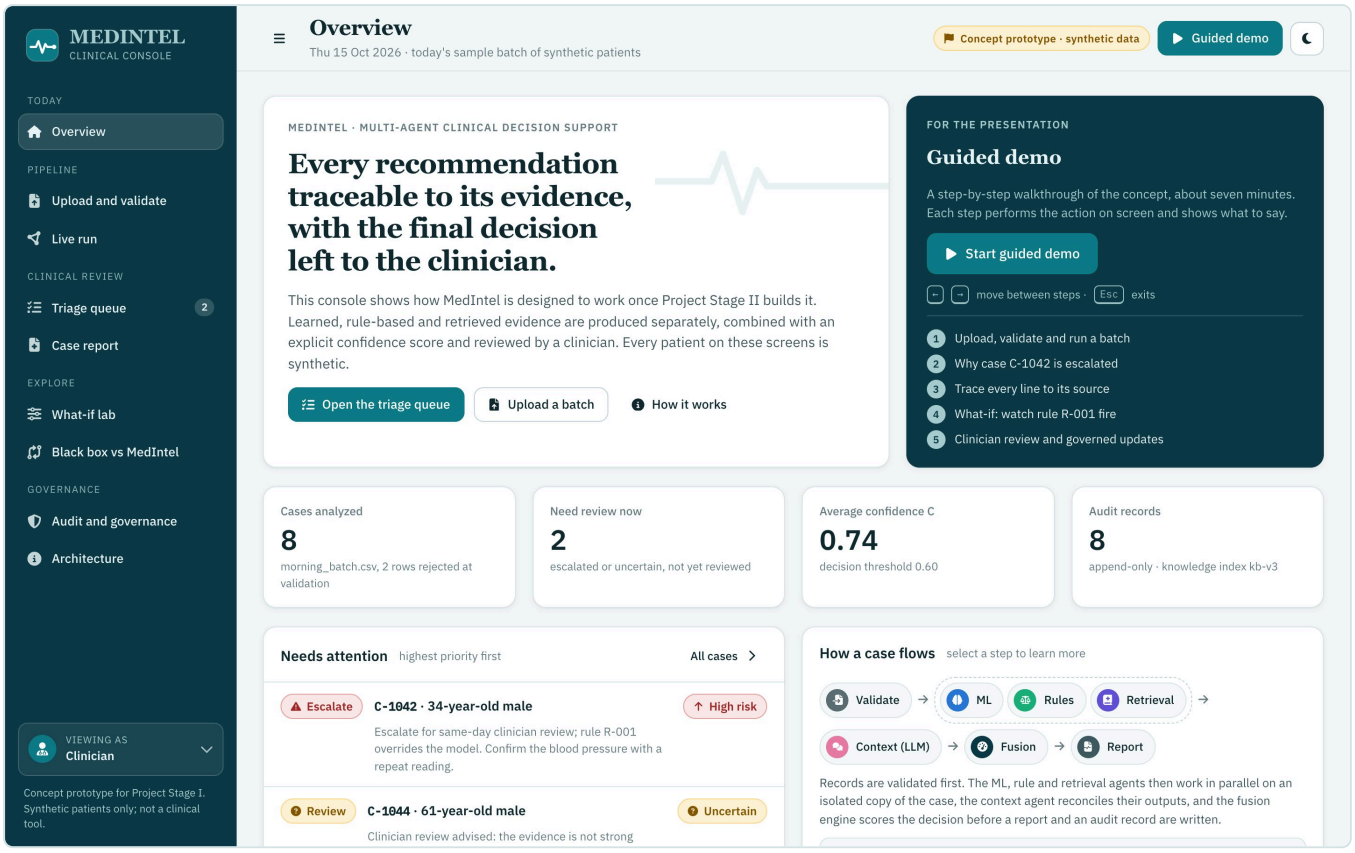


Figure 1. The Overview screen. The Guided demo card on the right starts the walkthrough, and the menu on the left opens each screen.

4 Presenting with the guided demo

The guided demo tells the whole story in 15 steps, about seven minutes. Each step performs the action for you (it uploads the batch, runs it, opens reports, moves sliders and switches roles), highlights where to look, and shows a card that explains it.

To	Press
Go to the next step	`→` or `Page Down`, the Next button, or a click on the dimmed area
Go back one step	`←` or `Page Up`
Leave the demo	`Esc` or the × on the card

A presentation clicker works too, because it sends the same Page Down and Page Up keys. On each card, the line with a teal bar is a suggested sentence for the presenter; say the rest in your own words.

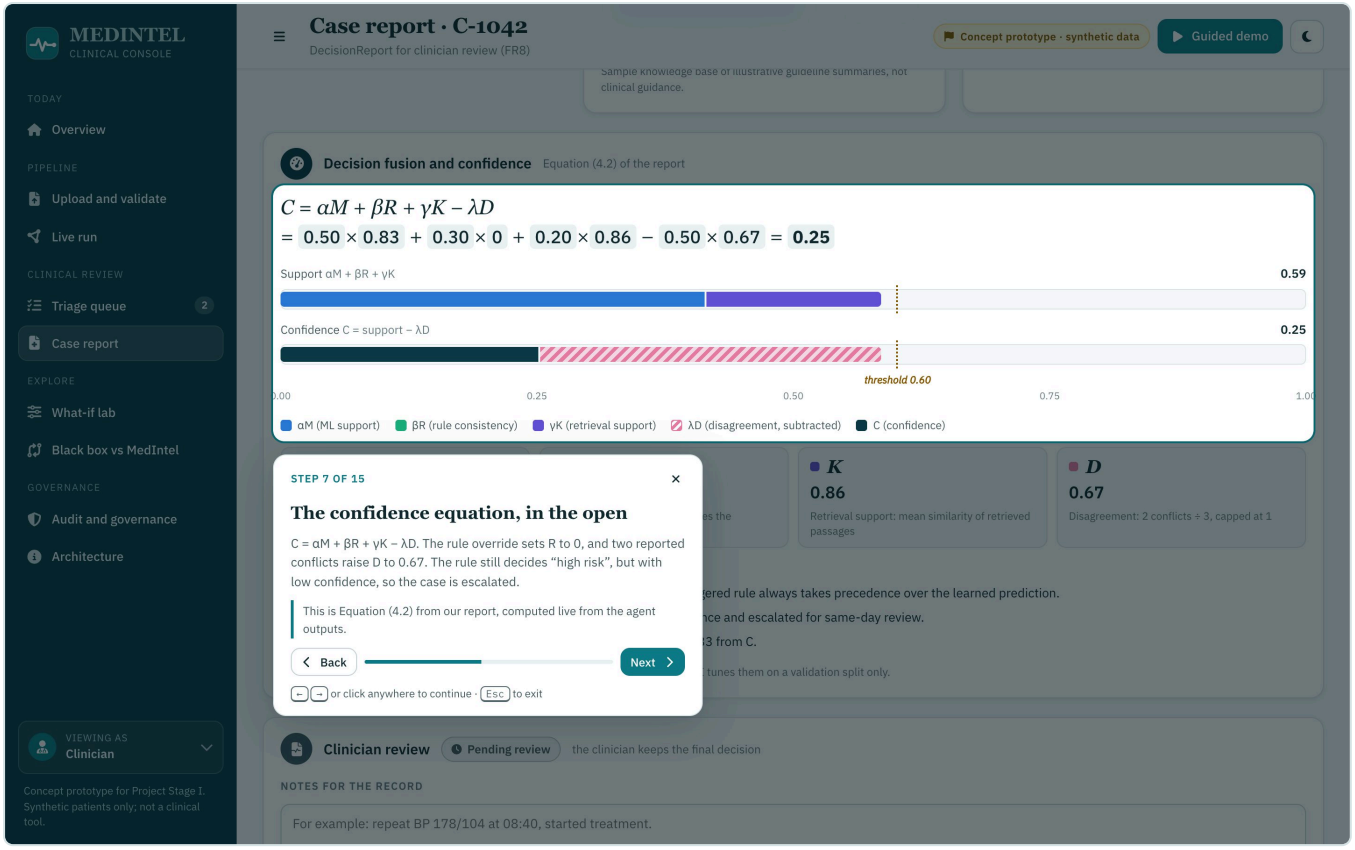


Figure 2. A guided demo step. The highlighted box shows where to look, and the card explains what is happening.

Demo script

Every step, in order, so you can rehearse. The suggested line is the sentence shown with a teal bar on the card.

Step	Card and screen	What happens	Suggested line
Intro	**MedIntel clinical console** Start	A concept prototype of how MedIntel is designed to work once Project Stage II builds it. Every patient is synthetic, and the steps below perform each action for you.	*MedIntel separates learned, rule-based and retrieved evidence, combines it with an explicit confidence score, and always leaves the decision to a clinician.*
1	**A morning batch arrives** Upload and validate	The Data Administrator uploads a de-identified CSV export. Here we use the sample batch of ten synthetic records.	*Records are only read in the browser; nothing leaves the machine.*
2	**Validation catches bad records first** Upload and validate	Ten records read, eight valid. One row has no patient ID (test case TC-01) and one has a temperature entered in °F (TC-02). Nothing is converted silently; both rows are held for correction.	*This is functional requirement FR1: no agent ever sees an invalid record.*
3	**Eight cases run in parallel, each in isolation** Live run	Four cases run at a time. Inside each case, the ML, rule and retrieval agents work in parallel; the context agent waits for all three, then fusion scores the decision.	*Each case has its own isolated context, so nothing from one patient can reach another patient's report (NFR2).*

4	**The triage queue puts risk first** Triage queue	Case C-1042 is at the top and marked Escalate: a safety rule overrides the model, and the confidence is low.	*The queue is ordered by priority, so the clinician sees the cases that need attention first.*
5	**A report built for review** Case report (C-1042)	High risk, with a confidence of 0.25, well below the 0.60 threshold. The recommendation says what to do next: same-day review and a repeat blood pressure reading.	*Explain the card in your own words.*
6	**Why: the model and the rules disagree** Case report (C-1042)	The ML model sees a 34-year-old non-smoker with normal labs and estimates a 17% risk. Rule R-001 fires on a systolic reading of 186 mmHg and elevates the case. The retrieved evidence on severely raised blood pressure sides with the rule.	*Each agent does one job and returns a typed result, so we can see exactly where they disagree.*
7	**The confidence equation, in the open** Case report (C-1042)	$C = \alpha M + \beta R + \gamma K - \lambda D$. The rule override sets R to 0, and two reported conflicts raise D to 0.67. The rule still decides “high risk”, but with low confidence, so the case is escalated.	*This is Equation (4.2) from our report, computed live from the agent outputs.*
8	**Every line is traceable** Case report (C-1042)	With Trace sources on, pointing at any part of the report lights up the component and version that produced it. Here the decision comes from the fusion engine, because of rule R-001.	*Explain the card in your own words.*
9	**Unsupported statements are removed** Case report (C-1047)	The language model’s draft claimed a history of diabetes. Nothing in the record supports it, so the sentence is struck out and kept out of the report (TC-12).	*The language model only summarizes; it is never allowed to add facts.*
10	**Saying “uncertain” instead of guessing** Case report (C-1044)	For C-1044 no sufficiently relevant evidence was found, so retrieval support is 0 and the confidence of 0.57 falls below the threshold. MedIntel reports “uncertain” and asks for a clinician’s view (TC-06).	*Explain the card in your own words.*
11	**What-if lab: watch a rule fire** What-if lab	We sweep C-1042’s systolic blood pressure from 170 to 190 mmHg. At 180, rule R-001 fires, a conflict appears and the confidence drops. Examiners can move any slider themselves.	*This is the boundary test from Chapter 6: 179, 180 and 181 mmHg.*
12	**The clinician makes the decision** Case report (C-1042)	The clinician accepts the escalation and adds a note. The review is appended to the audit trail, and the confirmed outcome is proposed to the Knowledge Curator.	*MedIntel recommends; it never acts on its own.*
13	**Knowledge changes only with approval** Audit and governance, knowledge updates	As Knowledge Curator we approve the validated outcome. The knowledge index moves from kb-v3 to kb-v4, and the change is recorded with who approved it.	*Model outputs never update the knowledge base by themselves, so one wrong answer cannot become a lasting error.*
14	**An append-only audit trail** Audit and governance, audit trail	Every report and every review is recorded with the model and rulebase versions and the retrieval IDs used, so any recommendation can be reproduced later (NFR1, NFR6).	*Explain the card in your own words.*

15	**Black box vs MedIntel** Black box vs MedIntel	The same case through a single opaque model: “low risk”, with no reasons, sources or record. MedIntel escalates it with evidence, conflicts, a traceable confidence and an audit record.	*This is the research gap from our literature survey, made visible.*
End	**Next: Project Stage II** Overview	Stage II builds the real system: a model trained on a public dataset, FAISS retrieval with sentence embeddings, a live language-model endpoint, and the twelve planned test cases.	*Thank you. We would be glad to take questions, and the console stays open for you to explore.*

Tips for the day

- Rehearse once on the laptop and projector you will use.
- If an examiner asks a question in the middle, press `Esc`, show the answer on screen, then click Guided demo again and press `→` until you are back at the same step. Each step sets up its own screen, so skipping ahead is safe.
- In a dim room, the moon button at the top right switches to a dark theme.

5 Exploring on your own

After the walkthrough the console stays open, and examiners can try it themselves. The menu on the left has nine screens.

Screen	What you can do
Overview	Today's batch at a glance and the cases that need attention first.
Upload and validate	Load a CSV batch: press Use the sample morning batch, or drag a file from sample_data onto the drop area. Rows with errors are shown in red with the reason and the test case number (TC-01, TC-02), and they never reach the agents.
Live run	Watch the cases run in parallel, four at a time. Change max_workers (1, 2, 4 or 8) and press Run again to see the difference.
Triage queue	All cases ordered by priority, with filters for cases that need review, pending cases and reviewed cases.
Case report	The full report for one case: patient snapshot, the four agent results, the fusion equation and the clinician review. Turn on Trace sources and point at any part of the report to see which component produced it. Export report shows the DecisionReport as JSON.
What-if lab	Change patient values or fusion weights and watch every agent respond (Section 6).
Black box vs MedIntel	The same case through a single opaque model and through MedIntel, side by side.
Audit and governance	The append-only audit trail, knowledge updates waiting for approval, the rulebase and the fusion settings.
Architecture	The seven components, what each one takes and produces, and what is computed live in this prototype.

Roles

The **Viewing as** button at the bottom of the left menu switches between the four roles from the report: Clinician, Data Administrator, Knowledge Curator and System Administrator. Some actions belong to one role: only a Data

Administrator uploads records, only a Clinician accepts or rejects a report, only a Knowledge Curator approves a knowledge update, and only a System Administrator changes the fusion settings. If you are in the wrong role, the screen shows a button that switches for you.

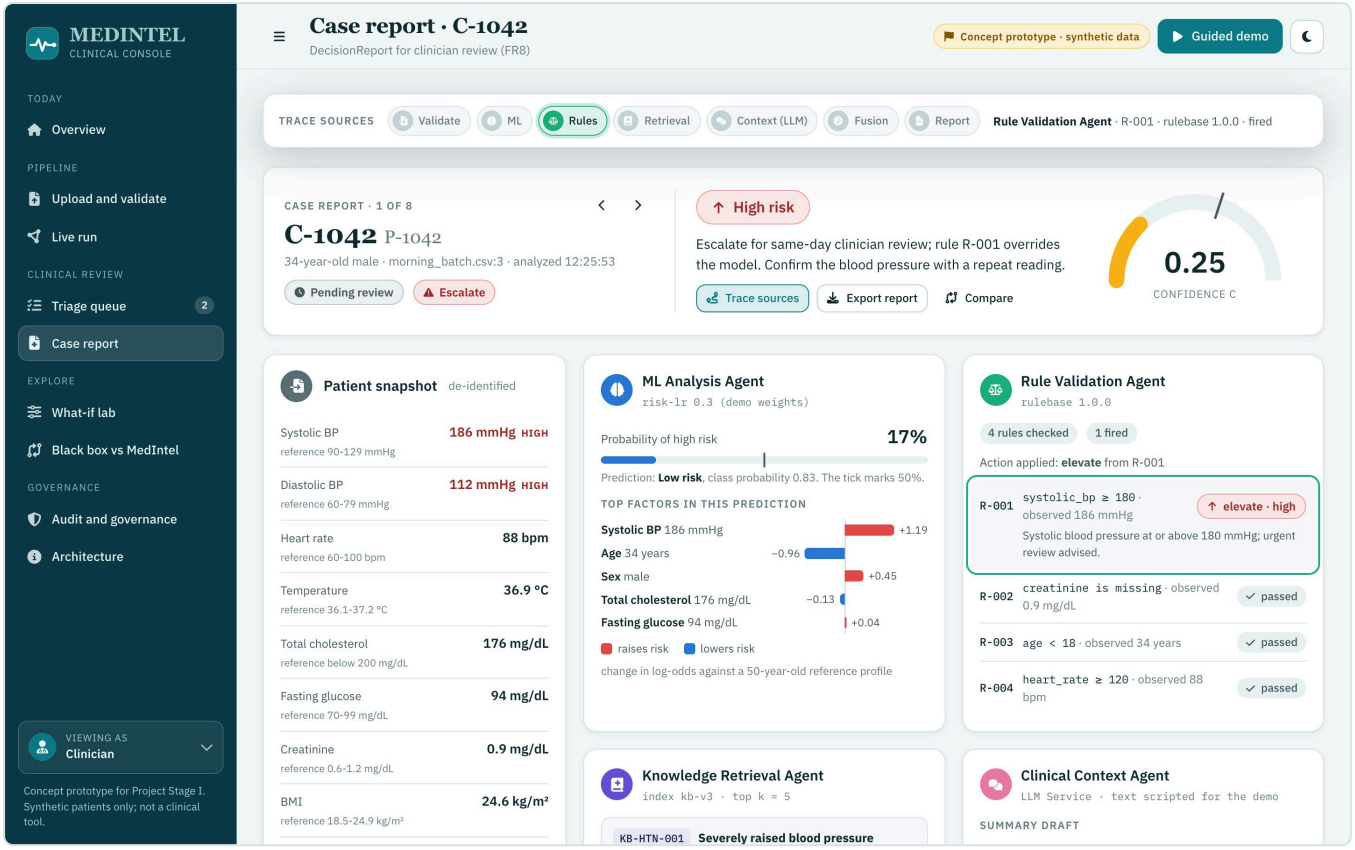


Figure 3. The case report for C-1042 with Trace sources on. Pointing at rule R-001 shows, in the bar at the top, that it came from the Rule Validation Agent and rulebase 1.0.0.

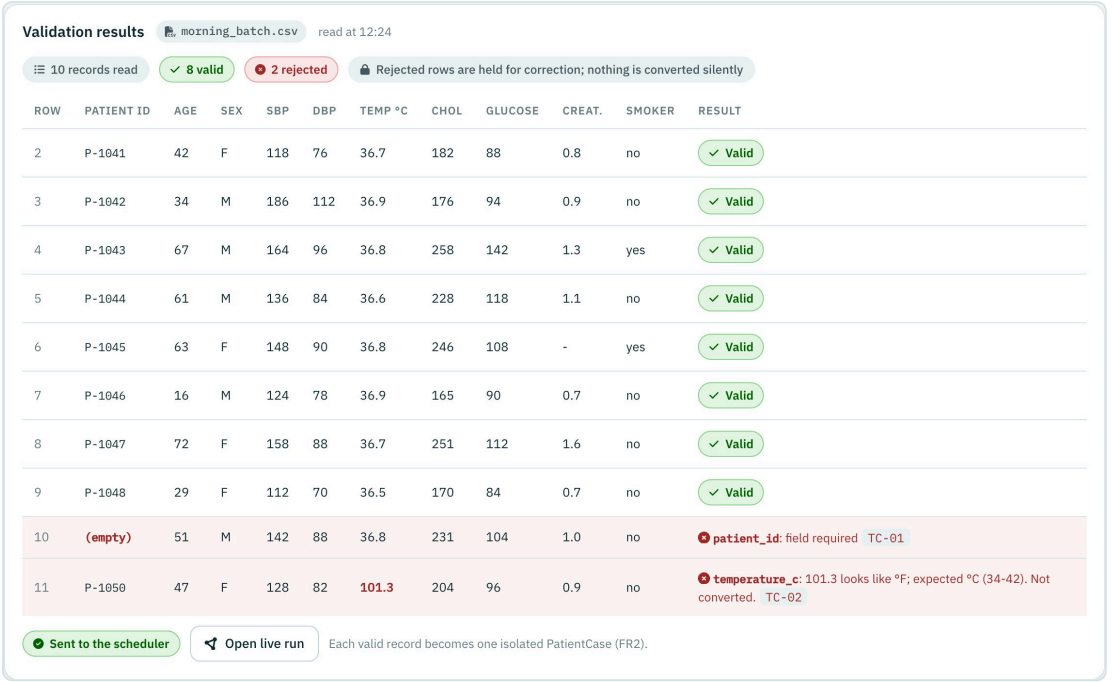


Figure 4. Validation results for the sample batch. Two rows are held back, each with the reason and its test case number.

6 Things to try

In the What-if lab

The lab starts from case C-1042. Every change is recomputed at once, and the What changed panel lists each effect in plain words.

- **Rule boundary:** move Systolic BP from 179 to 180. Rule R-001 fires, a conflict appears and the confidence C drops. The Sweep button does this for you, from 170 to 190.
- **Under 18:** set Age to 17. Rule R-003 withholds the recommendation.
- **Fast heart rate:** raise Heart rate to 120 or more. Rule R-004 adds a flag.
- **Missing lab result:** tick Creatinine result missing. Rule R-002 adds a flag, and the model uses a reference value instead.
- **Fusion weights:** α, β and γ always add up to 1. Raise λ to penalize disagreement more, or move the decision threshold and watch the decision become uncertain.
- Press **Reset** to go back to the original values.

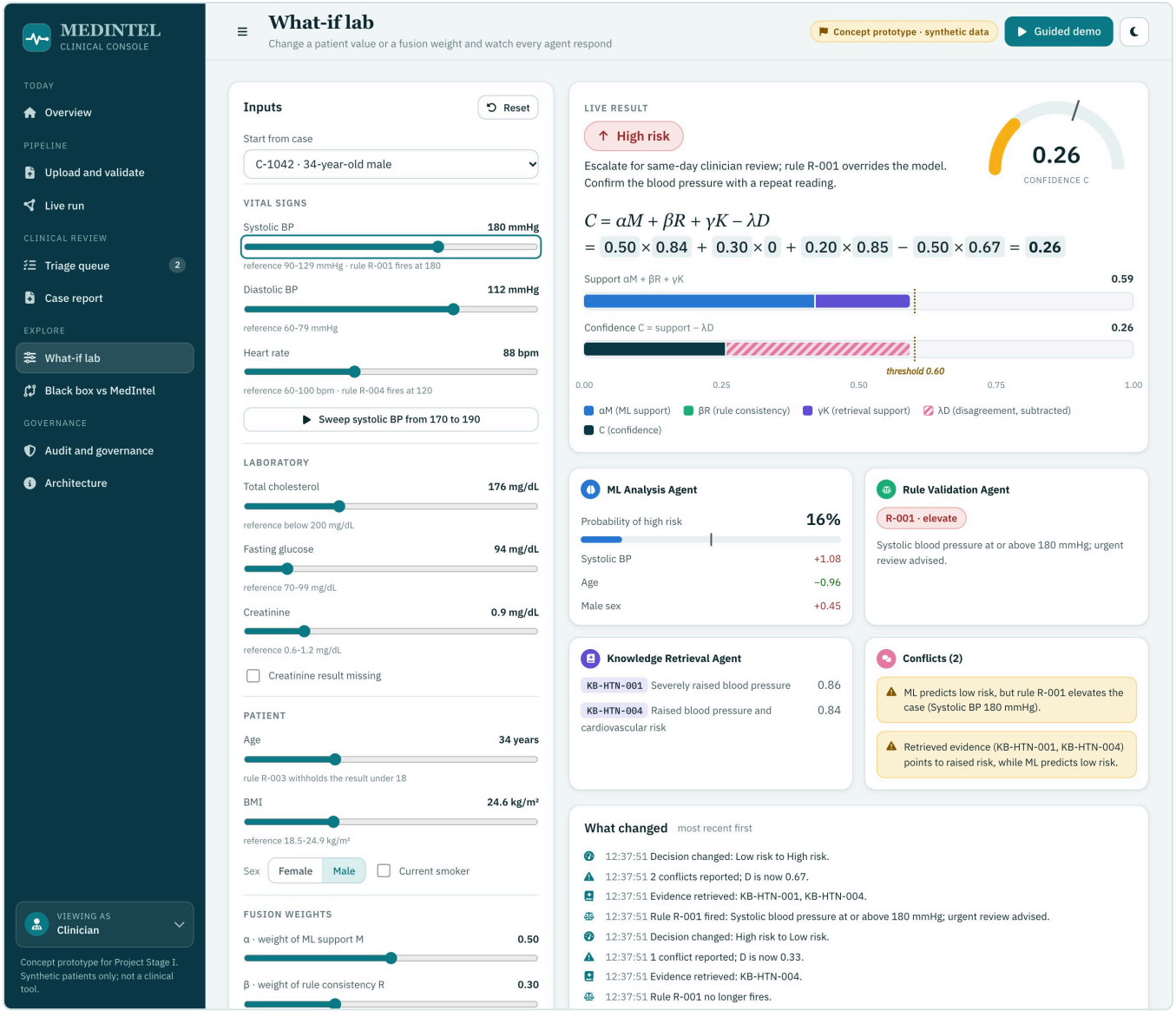


Figure 5. The What-if lab after systolic blood pressure was moved to 179 and then to 180 mmHg. The What changed panel at the bottom lists each effect.

With the boundary test file

On the Upload and validate screen (as Data Administrator), drag **sample_data/boundary_tests.csv** onto the drop area, or use Choose a file, and press Send to the scheduler. It shows the rules working on patients the prototype has never seen:

Patient	Input	Result
P-2001	Systolic 179 mmHg	No rule fires. Low risk; the evidence on raised blood pressure is reported as a conflict.
P-2002	Systolic 180 mmHg	R-001 fires. High risk, escalated.
P-2003	Systolic 181 mmHg	R-001 fires. High risk, escalated.
P-2004	Age 17	R-003 withholds the recommendation.
P-2005	Heart rate 124 bpm	R-004 adds a flag. Low risk.
P-2006	Creatinine missing	R-002 adds a flag. High risk.

To bring back the original patients afterwards, press Use the sample morning batch and send it again, or simply reload the page.

7 The sample patients

Case	Patient	What it shows
C-1042	34-year-old male, BP 186/112	The model estimates low risk (17%), but rule R-001 says high risk. MedIntel follows the rule, reports low confidence (C = 0.25) and escalates the case. This is the main story of the demo.
C-1044	61-year-old male	No sufficiently relevant evidence is found, so the decision is "uncertain" (TC-06).
C-1043	67-year-old male, smoker	High risk with high confidence (C = 0.92).
C-1047	72-year-old female	The language model's draft claims a history of diabetes that the record does not support, so the sentence is removed (TC-12).
C-1045	63-year-old female, smoker	Creatinine is missing, so rule R-002 adds a flag.
C-1046	16-year-old male	Under 18, so rule R-003 withholds the recommendation.
C-1041 C-1048	Healthy adults	Low risk with high confidence; routine follow-up.
Row 10	No patient ID	Rejected at validation (TC-01).
Row 11	P-1050, temperature 101.3	Looks like °F, so it is held for correction instead of being converted silently (TC-02).

8 What is real and what is simulated

Computed live in the browser

- CSV parsing and schema validation
- The risk estimate, from a logistic model with hand-set demo weights
- All four rules, with severity and precedence
- Evidence matching and similarity scores
- Conflict detection, Equation (4.2), rule precedence and the threshold
- Decision reports, audit records and approved knowledge updates

Pre-written or simulated

- Language-model summaries for the eight sample patients (other patients get a template summary)
- The knowledge passages: ten illustrative summaries, not clinical guidance
- The run timings on the Live run screen
- The single-model answer on the comparison screen
- All patients, notes and events

> If asked, a safe way to put it: "This is a concept prototype. The rules, the scoring and the fusion logic run live; the language-model text and the timings are simulated. Project Stage II replaces these parts with the real agents and a public dataset."

9 Questions examiners may ask

Is this the working system?

No. It is a concept prototype of the Stage I design. Validation, the rules, the risk score, evidence matching and the fusion step are computed live in the browser, while the language-model text is pre-written. Project Stage II builds the real agents.

Where does the confidence number come from?

From Equation (4.2) in our report, $C = \alpha M + \beta R + \gamma K - \lambda D$. The case report shows every term with its value, and the What-if lab lets you change them.

Why is C-1042 "high risk" but with low confidence?

Rule R-001 overrides the model, so the decision is high risk. Because the model and the rule disagree, R becomes 0 and the disagreement D rises, which lowers C to 0.25. A rule override with low confidence is escalated for same-day review.

Can the system change its own knowledge base?

No. A validated outcome is added only after a clinician accepts the report and a Knowledge Curator approves it. The index then moves from kb-v3 to kb-v4, and the audit trail records who approved it.

What stops the language model from making things up?

The Clinical Context Agent only summarizes the other agents' results. Any statement that the record does not support is removed before the report, as case C-1047 shows.

What happens to records with errors?

They are held back at validation with the reason, and no agent sees them (FR1). Nothing is converted silently.

Who makes the final decision?

Always the clinician. MedIntel recommends, and the Accept, Annotate and Reject buttons record the clinician's decision in the audit trail.

10 Troubleshooting

Problem	What to do
The page is blank	Extract the .zip first, then open the file in Chrome, Edge or Safari.
Text is too big or too small	Reset the zoom with `Ctrl` + `0` (`Cmd` + `0` on a Mac).
The projector cuts off the edges	Zoom out one step (`Ctrl` and minus) or lower the display resolution. The layout adapts to any screen size.
The guided demo is on the wrong screen	Press `Esc` and click Guided demo again. It starts from a clean state.
You want to start from scratch	Reload the page.
Saving an export	Export buttons show the JSON. Copy puts it on the clipboard; Save file downloads it to your Downloads folder.

11 Changing the prototype (optional)

The source_code folder holds everything used to build MedIntel_Prototype.html. You only need it to change text or behavior, for example during Project Stage II. With Node.js 18 or later installed:

1. Open a terminal in the source_code folder and run `npm install` once.
2. Edit the files in `src`. Patients, rules and knowledge passages are in `src/data.js`, the analysis logic is in `src/engine.js`, and the screens and the guided demo are in `src/app`.
3. Run `npm run build`. The new file is written to `dist/MedIntel_Prototype.html`.
4. Run `npm test` to check the expected result for each sample patient.

MedIntel is a design-stage concept for Project Stage I. This prototype uses synthetic data only. It is not a medical device and must not be used for real patient care.